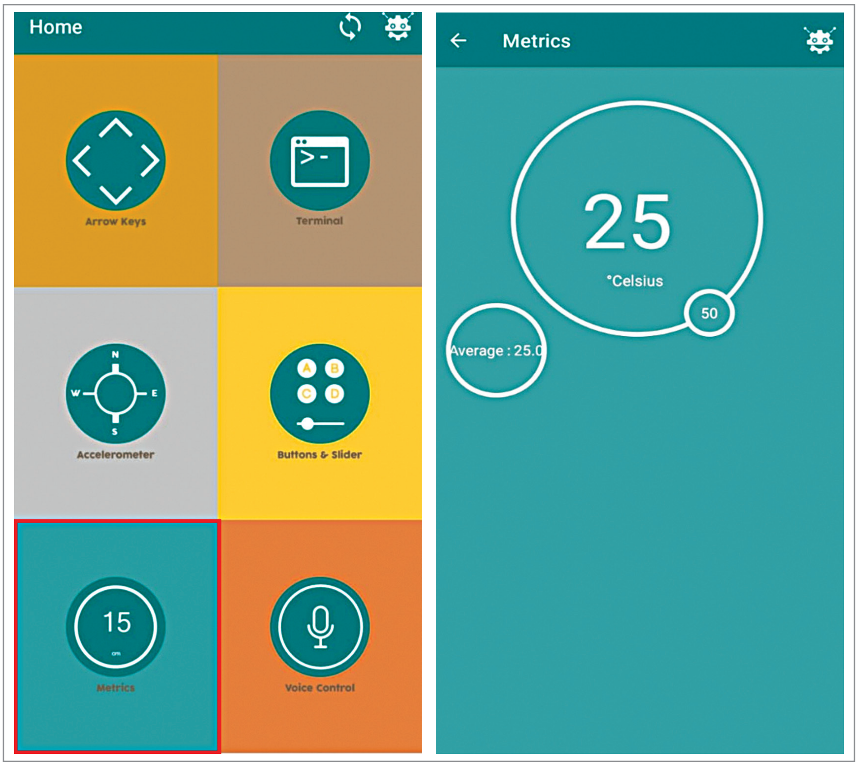

Fig. 9: App display

ject and select the port and board in Arduino IDE. Connect the ESP32 to your laptop and press the Boot button to bring the ESP32 board in program uploading mode and then upload the code to ESP32 board. The code can be download using the link *https://www.electronicsforu.com/electronics-projects/thermal-management-systems-for-electric-vehicle*

Construction and testing

After uploading the source code tms.ino into ESP32 board, connect the components as per circuit diagram in Fig. 3. In place of the voltage regulator circuit a 5V battery or 5V DC adaptor can be used to power the device. Relay connection prototype is shown in Fig. 5.

For installing the circuit into an electric vehicle refer Fig. 6. For testing the project refer Fig. 6 through Fig. 9. A Bluetooth terminal app is required to monitor the battery temperature and smoke data on phone. So, you may install the Arduino Blue Control app in your phone.

Power the device and connect the app with Bluetooth. You would now be able to see the temperature of battery in app. Whenever the temperature rises above the threshold set, or any smoke is detected by sensor, the app gives an alert and cuts off power supply to the vehicle's battery through the relay. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*


Both E. Venkatesan (left) and M. Dinesh (right) work as a Product Application Team Lead in Qmax Test Equipment Pvt Ltd and are interested in the areas of IoT, BMS, wireless technologies, and embedded systems.